Chapter 7. Principal Component Analysis

Population and sample principal components, SVD, choosing the number of components

Contents

Ch. 1, §6, introduced PCA through sample geometry. Here it is formulated at the population level, with the Lagrange-multiplier derivation, the SVD and the choice of the number of components.

1 Population Principal Components

Let $\mathbf{X} \in \mathbb{R}^p$ with $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$, $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$. A linear combination $Z(\mathbf{a}) = \mathbf{a}^\top \mathbf{X}$ has variance $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$. (For $d = 1$: $\mathbf{a}_1 = \arg\max_{\|\mathbf{a}\|=1} \mathrm{Var}(Z(\mathbf{a}))$.)

Definition 1: principal components

First PC direction vector: $\mathbf{u}_1 = \arg\max_{\|\mathbf{u}\|=1} \mathrm{Var}(Z(\mathbf{u}))$, with $\lambda_1 = \mathrm{Var}(Z_1)$. Second: $\mathbf{u}_2 = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_1 = 0} \mathrm{Var}(\mathbf{u}^\top \mathbf{X})$, with $\lambda_2 = \mathrm{Var}(Z_2)$; and so on:

$$\mathbf{u}_k = \arg \max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_j = 0\ (j<k)} \mathrm{Var}(\mathbf{u}^\top \mathbf{X}), \qquad Z_k = \mathbf{u}_k^\top \mathbf{X}.$$

Then $\lambda_1 \geq \lambda_2 \geq \cdots$ and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Theorem 2: solution

$\mathbf{u}_k$ is the kth eigenvector of $\boldsymbol{\Sigma}$, $\mathrm{Var}(Z_k) = \lambda_k$, and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Lagrange multipliers. **1.** $\Phi(\mathbf{u}, \lambda) = \mathrm{Var}(\mathbf{u}^\top \mathbf{X}) - \lambda(\mathbf{u}^\top \mathbf{u} - 1) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1)$. $\frac{1}{2} \frac{\partial}{\partial \mathbf{u}} \Phi(\mathbf{u}, \lambda) = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} = \mathbf{0}$, so $\boldsymbol{\Sigma}\mathbf{u} = \lambda \mathbf{u}$ and $\lambda = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}$: the objective equals the eigenvalue, so take the largest, λ_1.

2. With $\boldsymbol{\gamma}^k = (\gamma_1, \dots, \gamma_{k-1})$,

$$\Phi(\mathbf{u}, \lambda, \boldsymbol{\gamma}^k) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1) - \sum_{j=1}^{k-1} 2\gamma_j \mathbf{u}_j^\top \mathbf{u}, \qquad \frac{1}{2} \frac{\partial \Phi}{\partial \mathbf{u}} = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} - \sum_{j=1}^{k-1} \gamma_j \mathbf{u}_j = \mathbf{0}, \qquad \frac{\partial \Phi}{\partial \gamma_j} = \mathbf{u}_j^\top \mathbf{u} = 0.$$

Premultiplying by $\mathbf{u}_i^\top$ $(i < k)$: $\mathbf{u}_i^\top \boldsymbol{\Sigma} \mathbf{u} = \gamma_i$, and $\mathbf{u}_i^\top \boldsymbol{\Sigma} \mathbf{u} = \lambda_i \mathbf{u}_i^\top \mathbf{u} = 0$, so $\gamma_i = 0$. Hence $\boldsymbol{\Sigma}\mathbf{u} = \lambda \mathbf{u}$ again, and among eigenvectors orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$ the largest eigenvalue λ_k is chosen: $\mathbf{u}_k, \lambda_k$. Finally $\mathrm{Cov}(Z_j, Z_k) = \mathbf{u}_j^\top \boldsymbol{\Sigma} \mathbf{u}_k = \lambda_k \mathbf{u}_j^\top \mathbf{u}_k = 0$. ■

Theorem 3: total variance and correlations

1. $\sum_{j=1}^p \mathrm{Var}(X_j) = \mathrm{tr}\boldsymbol{\Sigma} = \sum_{k=1}^p \lambda_k = \sum_k \mathrm{Var}(Z_k)$; the proportion of variance explained by PC k is $\lambda_k / \sum_j \lambda_j$.

2. $\mathrm{Corr}(Z_k, X_j) = \dfrac{u_{jk}\sqrt{\lambda_k}}{\sqrt{\sigma_{jj}}}$, where u_{jk} is the jth entry of $\mathbf{u}_k$.

Proof. (1) $\mathrm{tr}(\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top) = \mathrm{tr}\boldsymbol{\Lambda}$. (2) $\mathrm{Cov}(Z_k, X_j) = \mathbf{u}_k^\top \boldsymbol{\Sigma} \mathbf{e}_j = \lambda_k \mathbf{u}_k^\top \mathbf{e}_j = \lambda_k u_{jk}$, so $\mathrm{Corr} = \lambda_k u_{jk} / (\sqrt{\lambda_k}\sqrt{\sigma_{jj}})$. ■

Theorem 4: Ky Fan maximum principle — q-dimensional subspaces

$$\max_{\mathbf{V} \in \mathbb{R}^{p \times q},\ \mathbf{V}^\top \mathbf{V} = \mathbf{I}_q} \mathrm{tr}(\mathbf{V}^\top \boldsymbol{\Sigma} \mathbf{V}) = \lambda_1 + \cdots + \lambda_q,$$

attained at $\mathbf{V} = [\mathbf{u}_1, \dots, \mathbf{u}_q]$ (and its rotations).

Proof. $\mathbf{P} = \mathbf{V}\mathbf{V}^\top$ is a rank-q projection with eigenvalues 1 (q times) and 0. By von Neumann's upper bound, $\operatorname{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V}) = \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{P}) \le \sum_i \lambda_i(\boldsymbol{\Sigma})\lambda_i(\mathbf{P}) = \lambda_1 + \cdots + \lambda_q$. ■

Remark 5: PCA as optimal approximation

$\mathrm{E}\|(\mathbf{X}-\boldsymbol{\mu}) - \mathbf{V}\mathbf{V}^\top(\mathbf{X}-\boldsymbol{\mu})\|^2 = \operatorname{tr}\boldsymbol{\Sigma} - \operatorname{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})$, so among q-dimensional subspaces through the mean, the first q PCs minimise the expected squared reconstruction error, the minimum being $\lambda_{q+1} + \cdots + \lambda_p$.

2 Sample Principal Components and the SVD

2.1 Sample PCs

Let $\tilde{\mathbf{X}} = \mathbf{H}_n\mathbf{X} = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top$ be the centred data matrix, $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\boldsymbol{\Lambda}}\hat{\mathbf{U}}^\top$, and

$$\hat{\mathbf{u}}_k = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top\hat{\mathbf{u}}_j=0,\ j=1,\dots,k-1} \mathbf{u}^\top\mathbf{S}\mathbf{u}, \qquad \text{scores } t_{ik} = \hat{\mathbf{u}}_k^\top(\mathbf{x}_i - \bar{\mathbf{x}}).$$

The sample PC scores have sample variance $\hat{\lambda}_k$ and are mutually uncorrelated.

2.2 Relation to the singular value decomposition

Write the SVD of the $p \times n$ matrix $\tilde{\mathbf{X}}^\top$ ($n > p$) as

$$\tilde{\mathbf{X}}^\top = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \underbrace{[\mathbf{u}_1 \cdots \mathbf{u}_p]}_{p\times p} \underbrace{\begin{pmatrix} d_1 & & \\ & \ddots & \\ & & d_p \end{pmatrix}}_{p\times p} \underbrace{\begin{pmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_p^\top \end{pmatrix}}_{p\times n}, \qquad \mathbf{U}^\top\mathbf{U} = \mathbf{I}_p,\ \mathbf{V}^\top\mathbf{V} = \mathbf{I}_p,\ d_1 \ge \cdots \ge d_p \ge 0.$$

Then

$$\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \mathbf{U}\operatorname{diag}\Big(\frac{d_1^2}{n-1}, \dots, \frac{d_p^2}{n-1}\Big)\mathbf{U}^\top, \qquad \begin{pmatrix} \mathbf{z}_{(1)}^\top \\ \vdots \\ \mathbf{z}_{(p)}^\top \end{pmatrix} = \mathbf{Z} = \mathbf{U}^\top\tilde{\mathbf{X}}^\top = \mathbf{D}\mathbf{V}^\top = \begin{pmatrix} d_1\mathbf{v}_1^\top \\ \vdots \\ d_p\mathbf{v}_p^\top \end{pmatrix}.$$

The PC directions are the left singular vectors, the eigenvalues are $d_k^2/(n-1)$, and the kth PC score vector is $\mathbf{z}_{(k)} = \hat{\mathbf{u}}_k^\top\tilde{\mathbf{X}}^\top = d_k\mathbf{v}_k^\top$. Numerically the SVD is preferred to forming $\mathbf{S}$ (computation of PCA).

3 PCA and the Best Approximating Affine Subspace

Theorem 6: claim

PCA with r components is equivalent to finding the best r-dimensional affine subspace (hyperplane) approximating the data.

3.1 The problem

Let the data be $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ and $r \in \{1, \dots, p\}$. We wish to approximate $\mathbf{x}_1, \dots, \mathbf{x}_n$ by $\bar{\mathbf{x}} + \mathbf{a}_1, \dots, \bar{\mathbf{x}} + \mathbf{a}_n$ where $\operatorname{rank}([\mathbf{a}_1, \dots, \mathbf{a}_n]) = r$. The approximation of $\mathbf{x}_i$ by $\bar{\mathbf{x}} + \mathbf{a}_i$ is better when the

error $\|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller, and the approximation of $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ by $\{\bar{\mathbf{x}} + \mathbf{a}_i\}$ is better when $\mathrm{SSE} = \sum_{i=1}^n \|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller. That is, we wish to find $\mathbf{A}_r = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ $(p \times n)$ solving

$$(\star) \qquad \min_{\mathbf{A}_r} \sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 \quad \text{subject to} \quad \operatorname{rank}(\mathbf{A}_r) = r.$$

Since every column of a rank-r matrix lies in an r-dimensional subspace, the points $\bar{\mathbf{x}} + \mathbf{a}_i$ lie on an r-dimensional affine subspace through $\bar{\mathbf{x}}$; conversely any such subspace gives feasible $\mathbf{A}_r$.

3.2 Observations

1. With $\tilde{\mathbf{X}}^\top = [\mathbf{x}_1 - \bar{\mathbf{x}}, \dots, \mathbf{x}_n - \bar{\mathbf{x}}]$ $(p \times n)$,

$$\sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 = \|\tilde{\mathbf{X}}^\top - \mathbf{A}_r\|_F^2,$$

because for a $p \times n$ matrix $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_n]$ the Frobenius norm is $\|\mathbf{B}\|_F^2 = \sum_{i=1}^p \sum_{j=1}^n b_{ij}^2 = \sum_{j=1}^n \|\mathbf{b}_j\|_2^2 = \operatorname{tr}(\mathbf{B}\mathbf{B}^\top) = \operatorname{tr}(\mathbf{B}^\top\mathbf{B})$.

2. The sample PCA of $(\mathbf{x}_1, \dots, \mathbf{x}_n)$ is given by

   (1) the eigendecomposition $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top = \sum_{i=1}^p \hat{\lambda}_i \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{\lambda}_i$ the ith PC variance, $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_r > \hat{\lambda}_{r+1} \geq \cdots \geq \hat{\lambda}_p$;

   (2) the SVD $\tilde{\mathbf{X}}_{(p\times n)}^\top = \hat{\mathbf{U}}\hat{\mathbf{D}}\hat{\mathbf{V}}^\top = \sum_{i=1}^p \hat{d}_i \hat{\mathbf{u}}_i \hat{\mathbf{v}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{d}_i\hat{\mathbf{v}}_i$ the ith PC score vector;

   (3) $\hat{\mathbf{u}}_i^\top \tilde{\mathbf{X}}^\top = \hat{d}_i \hat{\mathbf{v}}_i^\top$, i.e.

$$\begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix}.$$

Theorem 7: solution (Eckart–Young–Mirsky)

The $\mathbf{A}_r$ that minimises $(\star)$ is

$$\mathbf{A}_r = \sum_{i=1}^r \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r][\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]^\top \tilde{\mathbf{X}}^\top,$$

the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace, and the minimum is $\sum_{i>r} \hat{d}_i^2 = (n-1)\sum_{i=r+1}^p \hat{\lambda}_i$.

Proof. Step 1 (parametrise rank-r matrices). Since $\operatorname{rank}(\mathbf{A}_r) = r$, we can write $\mathbf{A}_r = \mathbf{E}\mathbf{C}^\top$ with $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_r]$ $(p \times r)$ satisfying $\mathbf{E}^\top\mathbf{E} = \mathbf{I}_r$ (an orthonormal basis of the column space of $\mathbf{A}_r$) and $\mathbf{C} = \begin{pmatrix} \mathbf{c}_1^\top \\ \vdots \\ \mathbf{c}_n^\top \end{pmatrix}$ an $n \times r$ matrix. Writing $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$, $(\star)$ is equivalent to $\min_{\mathbf{E},\mathbf{C}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2$, where

$$(\star\star) \qquad \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2 = \big\|(\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n) - \mathbf{E}[\mathbf{c}_1, \dots, \mathbf{c}_n]\big\|_F^2 = \sum_{i=1}^n \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2.$$

Step 2 (optimal $\mathbf{C}$ for fixed $\mathbf{E}$). For any given $\mathbf{E}$, $(\star\star)$ separates over i and its minimiser is

$$\hat{\mathbf{c}}_i = \arg\min_{\mathbf{c}_i} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2 = (\mathbf{E}^\top\mathbf{E})^{-1}\mathbf{E}^\top\tilde{\mathbf{x}}_i = \mathbf{E}^\top\tilde{\mathbf{x}}_i,$$

the projection scores of $\tilde{\mathbf{x}}_i$ onto span($\mathbf{E}$) (least squares). Stacking, $\hat{\mathbf{C}} = \begin{pmatrix} \tilde{\mathbf{x}}_1^\top \mathbf{E} \\ \vdots \\ \tilde{\mathbf{x}}_n^\top \mathbf{E} \end{pmatrix} = \tilde{\mathbf{X}}\mathbf{E}$, i.e. $\hat{\mathbf{C}}^\top = \mathbf{E}^\top \tilde{\mathbf{X}}^\top$.

Step 3 (profile out C). Plugging $\hat{\mathbf{C}}$ into ($\star\star$) (profiling, Theorem 6), and using that $\mathbf{I} - \mathbf{E}\mathbf{E}^\top$ is symmetric idempotent, $(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)^2 = \mathbf{I} - \mathbf{E}\mathbf{E}^\top$,

$$\begin{aligned} \min_{\mathbf{E}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{E}^\top \tilde{\mathbf{X}}^\top\|_F^2 &= \min_{\mathbf{E}} \|(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top\|_F^2 = \min_{\mathbf{E}} \operatorname{tr}(\tilde{\mathbf{X}}(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top) \\ &= \min_{\mathbf{E}} \Big\{ \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1)\operatorname{tr}\Big(\mathbf{E}^\top \frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}}{n-1} \mathbf{E}\Big) \Big\} \\ &= \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1) \max_{\mathbf{E}} \operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top). \end{aligned}$$

Step 4 (von Neumann). Recall $\mathbf{S} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top$. $\mathbf{E}\mathbf{E}^\top$ is symmetric with eigenvalues 1 (r times) and 0 ($p - r$ times), so von Neumann's trace inequality (Theorem 3) gives

$$\operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top) \le \hat{\lambda}_1 + \cdots + \hat{\lambda}_r,$$

with equality when $\mathbf{E} = [\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r]$ (or any $\mathbf{E}$ spanning the same space).

Step 5 (the minimiser). Therefore

$$\mathbf{A}_r = \mathbf{E}\hat{\mathbf{C}}^\top = [\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = [\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix} = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top$$

by Observation 2(3): the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace. Since $\operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) = (n-1)\operatorname{tr}\mathbf{S} = (n-1)\sum_i \hat{\lambda}_i$, the minimum is $(n-1)\sum_{i>r} \hat{\lambda}_i = \sum_{i>r} \hat{d}_i^2$. ■

Remark 8: interpretation

- The best approximating hyperplane is $\bar{\mathbf{x}} + \text{span}\{\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r\}$ and each $\mathbf{x}_i$ is approximated by its orthogonal projection $\bar{\mathbf{x}} + \sum_{k \le r} (\hat{\mathbf{u}}_k^\top \tilde{\mathbf{x}}_i)\hat{\mathbf{u}}_k$: the PC scores are the coordinates on the hyperplane.
- The fitted values satisfy $\hat{\mathbf{c}}_i = \mathbf{E}^\top \tilde{\mathbf{x}}_i$, i.e. reconstruction uses projection scores, exactly the decomposition $\mathbf{x} = \bar{\mathbf{x}} + \sum s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_j + r_{\mathcal{L}_q}(\mathbf{x})$ of Ch. 1.
- Minimising squared error and maximising retained variance (the $\arg\max \mathbf{u}^\top \mathbf{S}\mathbf{u}$ definition) are the same problem, since $\text{SSE} = (n-1)\{\operatorname{tr}\mathbf{S} - \operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top)\}$.
- If $\hat{\lambda}_r = \hat{\lambda}_{r+1}$ the optimal subspace is not unique; the strict gap $\hat{\lambda}_r > \hat{\lambda}_{r+1}$ makes it unique.

3.3 Correlation PCA

If the units differ, use PCA of standardised data: $\mathbf{R} = \mathbf{D}_{\mathbf{\Sigma}}^{-1/2} \mathbf{\Sigma} \mathbf{D}_{\mathbf{\Sigma}}^{-1/2} = \mathbf{U}_R \mathbf{\Lambda}_R \mathbf{U}_R^\top$ (sample: $\mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$). Since $\operatorname{tr}\mathbf{R} = p$, the proportion explained is λ_k^R / p. Covariance and correlation PCs are in general not simple transformations of each other (scale dependence).

4 How Many Components to Keep?

- **Scree plot / cumulative scree plot**: plot $\hat{\lambda}_k$ against k and look for the elbow, or take the q at which the cumulative proportion reaches a target (e.g. 80–90%).

- **Kaiser's rule**: keep components with $\hat{\lambda}_k > \bar{\lambda} = \frac{1}{p}\sum_j \hat{\lambda}_j$; for correlation PCA, $\hat{\lambda}_k > 1$.
- **Likelihood ratio testing**: under normality test $H_0(k) : \lambda_{k+1} = \cdots = \lambda_p$ (the remaining space is spherical). With arithmetic mean $\bar{a}$ and geometric mean $\bar{g}$ of the last $p-k$ sample eigenvalues,

$$-2\log\Lambda = n(p-k)\log\frac{\bar{a}}{\bar{g}} \;\dot{\sim}\; \chi^2_{(p-k+2)(p-k-1)/2};$$

since $\bar{a} \geq \bar{g}$ (AM–GM), the statistic is small when the eigenvalues are nearly equal.

Remark 9: cautions in interpretation

- The sign of an eigenvector is arbitrary; what matters is the sign pattern (same signs: "size"; mixed signs: "contrast"). In the scores example $\mathbf{u}_1 \propto (1,1)$ was overall level and $\mathbf{u}_2 \propto (1,-1)$ the contrast between subjects.
- With tied eigenvalues the directions within the eigenspace are not unique.
- Sample eigenvalues are biased: large ones are overestimated and small ones underestimated. Under normality, for distinct λ_k, $\sqrt{n}(\hat{\lambda}_k - \lambda_k) \Rightarrow N(0, 2\lambda_k^2)$.
- High explained variance does not mean usefulness for classification or prediction (the $X_1 = 10Z$, $X_2 = Y$ example of Ch. 1).

5 Example: Scores in Two Subjects Revisited

For $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$ the characteristic equation $(125-\lambda)^2 - 75^2 = 0$ gives $\lambda = 200, 50$; the null space of $\mathbf{S} - 200\mathbf{I} = \begin{pmatrix} -75 & 75 \\ 75 & -75 \end{pmatrix}$ is spanned by $(1,1)^\top$. $\mathrm{Corr}(Z_1, X_1) = u_{11}\sqrt{\lambda_1}/\sqrt{s_{11}} = \frac{1}{\sqrt{2}}\sqrt{200}/\sqrt{125} = 0.894$, $\mathrm{Corr}(Z_2, X_1) = \frac{1}{\sqrt{2}}\sqrt{50}/\sqrt{125} = 0.447$, and $0.894^2 + 0.447^2 = 1$: each variable's variance is fully decomposed over the PCs.